

LIMITS OF EXPONENTIAL FUNCTIONS

Basic Calculus



Limit of the General Exponential Function $y = b^x$

- For all real numbers c , $\lim_{x \rightarrow c} b^x = b^c$.
- If $b > 1$, then $\lim_{x \rightarrow \infty} b^x = \infty$ and $\lim_{x \rightarrow -\infty} b^x = 0$.
- If $0 < b < 1$, then $\lim_{x \rightarrow \infty} b^x = 0$ and $\lim_{x \rightarrow -\infty} b^x = \infty$.

EXAMPLE 1: EVALUATE THE FOLLOWING LIMITS:

a. $\lim_{x \rightarrow 3} 2^x$

b. $\lim_{x \rightarrow -5} 2^x$

c. $\lim_{x \rightarrow \infty} 2^x$

d. $\lim_{x \rightarrow -\infty} 2^x$

SOLUTIONS

- By direct substitution, $\lim_{x \rightarrow 3} 2^x = 2^3 = 8$.
- By direct substitution, $\lim_{x \rightarrow -5} 2^x = 2^{-5} = \frac{1}{2^5} = \frac{1}{32}$.
- Because $b = 2$ and $b > 1$, then $\lim_{x \rightarrow \infty} 2^x = \infty$.
- Because $b > 1$, then $\lim_{x \rightarrow -\infty} 2^x = 0$.

**EXAMPLE 2: EVALUATE THE
FOLLOWING:**

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$$\lim_{x \rightarrow \infty} \left(\frac{1}{2} \right)^{x+2}$$

SOLUTION:

- Simplify the function first to get the basic exponential expression b^x . Use the product rule of the laws of exponents.

$$\begin{aligned}\left(\frac{1}{2}\right)^{x+2} &= \left(\frac{1}{2}\right)^x \cdot \left(\frac{1}{2}\right)^2 \\ &= \left(\frac{1}{2}\right)^x \cdot \frac{1}{4}\end{aligned}$$

SOLUTION:

•Thus,

$$\begin{aligned}
 \lim_{x \rightarrow \infty} x^{\frac{1}{x} + 2} &= \lim_{x \rightarrow \infty} x^{\frac{1}{x}} x^2 \\
 &= \lim_{x \rightarrow \infty} x^{\frac{1}{x}} \lim_{x \rightarrow \infty} x^2 \\
 &= \lim_{x \rightarrow \infty} x^{\frac{1}{x}} \cdot \frac{1}{4} \\
 &= 0 \cdot \frac{1}{4} = 0
 \end{aligned}$$

Substitution

Product rule

Constant rule

Limit of $x^{\frac{1}{x}}$ when $x < 1$

**EXAMPLE 3: EVALUATE THE
FOLLOWING:**

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$$\lim_{x \rightarrow -\infty} 4^{-x+3}$$

SOLUTION:

- Use the laws of exponents to simplify 4^{-x+3} .

$$\begin{aligned}4^{-x+3} &= 4^{-x} \cdot 4^3 \\&= (4^{-1})^x \cdot 64 \\&= \left(\frac{1}{4}\right)^x \cdot 64\end{aligned}$$

SOLUTION:

• Therefore,

$$\begin{aligned}\lim_{x \rightarrow -\infty} 4^{-x+3} &= \lim_{x \rightarrow -\infty} \left[\left(\frac{1}{4} \right)^x \cdot 64 \right] \\ &= \lim_{x \rightarrow -\infty} \left(\frac{1}{4} \right)^x \cdot \lim_{x \rightarrow -\infty} 64 \\ &= \infty \cdot 64 \\ &= \infty\end{aligned}$$

**EXAMPLE 4: EVALUATE THE
FOLLOWING:**

$$\lim_{x \rightarrow \infty} \frac{4^x + 3^x}{2^x}$$

SOLUTION:

- Apply the limit laws to solve the problem.

$$\lim_{x \rightarrow \infty} \frac{4^x + 3^x}{2^x} = \frac{\lim_{x \rightarrow \infty} 4^x + \lim_{x \rightarrow \infty} 3^x}{\lim_{x \rightarrow \infty} 2^x} = \frac{\infty}{\infty}$$

- Note that this is an indeterminate form. To circumvent this form, divide each term by the highest power (4^x).

SOLUTION: 

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$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{4^x + 3^x}{2^x} &= \lim_{x \rightarrow \infty} \frac{\frac{4^x}{4^x} + \frac{3^x}{4^x}}{\frac{2^x}{4^x}} \\ &= \lim_{x \rightarrow \infty} \frac{1 + \left(\frac{3}{4}\right)^x}{\left(\frac{1}{2}\right)^x}\end{aligned}$$

SOLUTION:

- $$= \frac{\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \left(\frac{3}{4}\right)^x}{\lim_{x \rightarrow \infty} \left(\frac{1}{2}\right)^x}$$
$$= \frac{1 + 0}{0} = \frac{1}{0} = \infty$$

**EXAMPLE 5: EVALUATE THE
FOLLOWING:**

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$$\lim_{x \rightarrow -\infty} \frac{e^{4x} + 2}{e^x + e^{2x}}$$

SOLUTION:

- $$\begin{aligned}
 \lim_{x \rightarrow -\infty} \frac{e^{4x} + 2}{e^x + e^{2x}} &= \frac{\lim_{x \rightarrow -\infty} e^{4x} + \lim_{x \rightarrow -\infty} 2}{\lim_{x \rightarrow -\infty} e^x + \lim_{x \rightarrow -\infty} e^{2x}} \\
 &= \frac{\lim_{x \rightarrow -\infty} (e^4)^x + \lim_{x \rightarrow -\infty} 2}{\lim_{x \rightarrow -\infty} e^x + \lim_{x \rightarrow -\infty} (e^2)^x} \\
 &= \frac{0 + 2}{0 + 0} = \frac{2}{0} = \infty
 \end{aligned}$$

- Note: because e^4 and e^2 are greater than

EXAMPLE 6: EVALUATE THE
FOLLOWING:

$$\lim_{x \rightarrow \infty} \frac{3 + 2e^{3x}}{5^x}$$

SOLUTION:

- Apply the limit laws to solve the problem.

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{3 + 2e^{3x}}{5^x} &= \frac{\lim_{x \rightarrow \infty} 3 + 2 \lim_{x \rightarrow \infty} e^{3x}}{\lim_{x \rightarrow \infty} 5^x} \\
 &= \frac{\lim_{x \rightarrow \infty} 3 + 2 \lim_{x \rightarrow \infty} (e^3)^x}{\lim_{x \rightarrow \infty} 5^x} \\
 &= \frac{3 + \infty}{\infty} = \frac{\infty}{\infty}
 \end{aligned}$$

SOLUTION:

- To circumvent the indeterminate form, divide each term by e^{3x} .

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{3 + 2e^{3x}}{5^x} &= \lim_{x \rightarrow \infty} \frac{\frac{3}{e^{3x}} + \frac{2e^{3x}}{e^{3x}}}{\frac{5^x}{e^{3x}}} \\
 &= \frac{3 \lim_{x \rightarrow \infty} \left(\frac{1}{e^3}\right)^x + \lim_{x \rightarrow \infty} 2}{\lim_{x \rightarrow \infty} \left(\frac{5}{e^3}\right)^x} \\
 &= \frac{3(0) + 2}{0} = \frac{2}{0} = \infty
 \end{aligned}$$

EVALUATE THE
FOLLOWING:

- $\lim_{x \rightarrow \infty} \frac{5e^x + e^{2x}}{e^{3x} + e^x}$

- $\lim_{x \rightarrow \infty} \frac{2^x + 5^x}{3^x}$

LIMITS OF LOGARITHMIC FUNCTIONS

Basic Calculus



Limit of $y = \log_b x$

- For all real numbers c in the domain of $y = \log_b x$,
 $\lim_{x \rightarrow c} [\log_b x] = \log_b c$.
- If $b > 1$, then $\lim_{x \rightarrow \infty} \log_b x = \infty$ and $\lim_{x \rightarrow 0^+} \log_b x = -\infty$.
- If $0 < b < 1$,
then $\lim_{x \rightarrow \infty} \log_b x = -\infty$ and $\lim_{x \rightarrow 0^+} \log_b x = \infty$.

EXAMPLE 8: EVALUATE THE FOLLOWING LIMITS:

a. $\lim_{x \rightarrow 32} [\log_2 x]$

b. $\lim_{x \rightarrow \frac{1}{6}} [\log_3 2x]$

c. $\lim_{x \rightarrow \infty} [\log_{\frac{1}{3}} 9x^3]$

SOLUTIONS:

- By direct substitution of $x = 32$,
$$\lim_{x \rightarrow 32} [\log_2 x] = \log_2 32 = 5.$$
- Directly substituting $x = \frac{1}{6}$, $\lim_{x \rightarrow \frac{1}{6}} [\log_3 2x] =$
$$\log_3 2 \left(\frac{1}{6} \right) = \log_3 \frac{1}{3} = -1.$$

SOLUTIONS:

- Using the laws of logarithms, simplify first the given logarithmic expression.

$$\begin{aligned}
 \log_{\frac{1}{3}} 9 \times 3 &= \log_{\frac{1}{3}} 9 + \log_{\frac{1}{3}} 3 \\
 &= \log_{\frac{1}{3}} 9 + 3 \log_{\frac{1}{3}} 3 \\
 &= -2 + 3 \log_{\frac{1}{3}} 3
 \end{aligned}$$

Product rule of logarithms

Power rule of logarithms

$$\log_{\frac{1}{3}} 9 = -2 \text{ (definition of logarithms)}$$

SOLUTIONS:

•Therefore,

$$\begin{aligned}
 \lim_{x \rightarrow \infty} x \log_{\frac{1}{3}} 9x^3 &= \lim_{x \rightarrow \infty} x(-2 + 3 \log_{\frac{1}{3}} x) \\
 &= \lim_{x \rightarrow \infty} -2 + \lim_{x \rightarrow \infty} 3 \log_{\frac{1}{3}} x \\
 &= -2 + 3 \lim_{x \rightarrow \infty} \log_{\frac{1}{3}} x \\
 &= -2 + 3(-\infty) \\
 &= -\infty
 \end{aligned}$$

Substitution

Sum rule

Constant and constant
multiple rule

$$\lim_{x \rightarrow \infty} \log_{\frac{1}{3}} x = -\infty \text{ when } 0 < \frac{1}{3} < 1$$

**EXAMPLE 9: EVALUATE THE
FOLLOWING:**

- $$\lim_{x \rightarrow 0^+} [\ln(4x)]$$

SOLUTION:

- Recall that $\ln(4x)$ means the natural logarithm of $4x$. A natural logarithm has a base of e , which is an irrational number equivalent to 2.718281828285 Thus, in a natural logarithm, $b > 1$.

SOLUTION:

- $$\begin{aligned}\lim_{x \rightarrow 0^+} [\ln(4x)] &= \lim_{x \rightarrow 0^+} [\ln 4 + \ln x] \\ &= \lim_{x \rightarrow 0^+} [\ln 4] \\ &= \ln 4 + (-\infty) \\ &= -\infty\end{aligned}$$

LIMITS OF TRIGONOMETRIC FUNCTIONS

Basic Calculus



LIMIT OF $\sin x$ AND $\cos x$

- Because $y = \sin x$ and $y = \cos x$ are continuous functions, which means they are defined for any real number x , then the limit as x approaches a certain number may be evaluated using direct substitution. In symbols,

$$\lim_{x \rightarrow c} (\sin x) = \sin c$$
$$\lim_{x \rightarrow c} (\cos x) = \cos c$$

EXAMPLE:

- Without using table of values and graph, evaluate

$$\lim_{x \rightarrow \frac{\pi}{4}} (\cos 2x)$$

SOLUTION:

- By direct substitution, you have

$$\begin{aligned}\lim_{x \rightarrow \frac{\pi}{4}} (\cos 2x) &= \cos 2 \left(\frac{\pi}{4} \right) \\ &= \cos \left(\frac{\pi}{2} \right) \\ &= 0\end{aligned}$$

EXAMPLE:

• Evaluate $\lim_{x \rightarrow \frac{\pi}{3}} \left(\sin \frac{x}{4} \right)$

SOLUTION:

Direct substitution

Simplification

Difference of two angles

Difference identity for sine

$$\begin{aligned}
 \lim_{x \rightarrow \frac{\pi}{3}} \sin \frac{x}{4} &= \sin \frac{\frac{\pi}{3}}{4} \\
 &= \sin \frac{\pi}{12} \\
 &= \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\
 &= \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} \\
 &= \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) - \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = \frac{\sqrt{6} - \sqrt{2}}{4}
 \end{aligned}$$

EXAMPLE:

- Evaluate $\lim_{x \rightarrow \frac{\pi}{4}} (\tan 3x)$

SOLUTION:

- Based on the fundamental identities,
 $\tan 3x = \frac{\sin 3x}{\cos 3x}$. Thus,

$$\begin{aligned}
 \lim_{x \rightarrow \frac{\pi}{4}} (\tan 3x) &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin 3x}{\cos 3x} \\
 &= \frac{\lim_{x \rightarrow \frac{\pi}{4}} (\sin 3x)}{\lim_{x \rightarrow \frac{\pi}{4}} (\cos 3x)} \\
 &= \frac{\sin \frac{3\pi}{4}}{\cos \frac{3\pi}{4}} \\
 &= \frac{\frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2}} = -1
 \end{aligned}$$

SOLUTION:

Quotient/Ratio identity

Quotient rule of Limit laws

Direct substitution

THE VALUE OF $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ AND $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

- In terms of trigonometric functions, two limits that are of interest are $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$. This is because the direct substitution method will not apply to the indicated trigonometric expressions, as doing so results in having an indeterminate number.

THE VALUE OF $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ AND $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

For all x in the domain $y = \frac{\sin x}{x}$ and $y = \frac{1 - \cos x}{x}$

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$

EXAMPLE:

• Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

SOLUTION:

- By direct substitution,

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \frac{\sin 3(0)}{0} = \frac{\sin 0}{0} = \frac{0}{0}$$

- Let $t = 3x$ or $x = \frac{t}{3}$. If x approaches 0, then $t = 3x$ also approaches 0. Thus, you have

SOLUTION:

- $$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 3x}{x} &= \lim_{t \rightarrow 0} \frac{\sin t}{\frac{t}{3}} \\ &= \lim_{t \rightarrow 0} \left[3 \cdot \frac{\sin t}{t} \right] \\ &= \lim_{t \rightarrow 0} 3 \cdot \lim_{t \rightarrow 0} \frac{\sin t}{t} \\ &= 3 \cdot 1 \\ &= 3\end{aligned}$$

EXAMPLE:

- Find the limit of $\lim_{x \rightarrow 0} \frac{1 - \cos 5x}{6x}$

SOLUTION:

- Using direct substitution,

$$\lim_{x \rightarrow 0} \frac{1 - \cos 5x}{6x} = \frac{1 - \cos 0}{0} = \frac{0}{0}$$

- Let $t = 5x$ or $x = \frac{t}{5}$. If x approaches 0, then $t = 5x$ approaches 0. Thus, you have

SOLUTION:

$$\begin{aligned}
 \bullet \quad \lim_{x \rightarrow 0} \frac{1 - \cos 5x}{6x} &= \lim_{t \rightarrow 0} \frac{1 - \cos t}{\frac{6t}{5}} \\
 &= \lim_{t \rightarrow 0} \left[\frac{5}{6} \cdot \frac{1 - \cos t}{t} \right] \\
 &= \lim_{t \rightarrow 0} \frac{5}{6} \cdot \lim_{t \rightarrow 0} \frac{1 - \cos t}{t} \\
 &= \frac{5}{6} \cdot 0 \\
 &= 0
 \end{aligned}$$

EXAMPLE:

• Evaluate $\lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^2-1}$

SOLUTION:

- By direct substitution,

$$\lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^2-1} = \frac{\sin 0}{0} = \frac{0}{0}$$

- You know that

$$\frac{\sin(x-1)}{x^2-1} = \frac{\sin(x-1)}{(x+1)(x-1)}$$

- Therefore, let $t = x - 1$ or $x = t + 1$.
If x approaches 1, then $t = x - 1$ approaches 0. Therefore, you have

SOLUTION:

- $$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^2-1} &= \lim_{t \rightarrow 0} \frac{\sin(x-1)}{(x+1)(x-1)} \\ &= \lim_{t \rightarrow 0} \frac{\sin t}{(t+1+1)(t+1-1)} \\ &= \lim_{t \rightarrow 0} \left[\frac{1}{t+2} \cdot \frac{\sin t}{t} \right] \end{aligned}$$

SOLUTION:

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$$\begin{aligned} &= \lim_{t \rightarrow 0} \frac{1}{t + 2} \cdot \lim_{t \rightarrow 0} \frac{\sin t}{t} \\ &= \frac{1}{0 + 2} \cdot 1 \\ &= \frac{1}{2} \end{aligned}$$

EXAMPLE:

• Evaluate $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan 3x}{x - \frac{\pi}{3}}$

SOLUTION:

- By direct substitution,

$$\begin{aligned}\lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan 3x}{x - \frac{\pi}{3}} &= \frac{\tan 3\left(\frac{\pi}{3}\right)}{\frac{\pi}{3} - \frac{\pi}{3}} \\ &= \frac{\tan \pi}{0} = \frac{0}{0}\end{aligned}$$

- To circumvent the independent form, let $t = x - \frac{\pi}{3}$, or equivalently, $x = t + \frac{\pi}{3}$. Note that as x approaches $\frac{\pi}{3}$, t approaches 0 so that

SOLUTION:

$$\begin{aligned}
 \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan 3x}{x - \frac{\pi}{3}} &= \lim_{t \rightarrow 0} \frac{\tan 3 \left(t + \frac{\pi}{3} \right)}{t} \\
 &= \lim_{t \rightarrow 0} \frac{\tan(3t + \pi)}{\tan 3t + \tan \pi} \\
 &= \lim_{t \rightarrow 0} \frac{1 - \tan 3t \tan \pi}{t} \\
 &= \lim_{t \rightarrow 0} \left[\frac{1}{t} \cdot \frac{\tan 3t + \tan \pi}{1 - \tan 3t \tan \pi} \right]
 \end{aligned}$$

SOLUTION:

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$$\begin{aligned}
 &= \lim_{t \rightarrow 0} \left[\frac{1}{t} \cdot \frac{\tan 3t + 0}{1 - \tan 3t(0)} \right] \\
 &= \lim_{t \rightarrow 0} \left[\frac{1}{t} \cdot (\tan 3t) \right] \\
 &= \lim_{t \rightarrow 0} \left[\frac{1}{t} \cdot \frac{\sin 3t}{\cos 3t} \right] \\
 &= \lim_{t \rightarrow 0} \left[\frac{1}{\cos 3t} \cdot \frac{\sin 3t}{t} \right]
 \end{aligned}$$

SOLUTION:

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$$\begin{aligned} &= \lim_{t \rightarrow 0} \frac{1}{\cos 3t} \cdot \lim_{t \rightarrow 0} \frac{\sin 3t}{t} \\ &= \frac{1}{\cos 0} \cdot \lim_{t \rightarrow 0} \frac{\sin 3t}{t} \\ &= 1 \cdot \lim_{t \rightarrow 0} \frac{\sin 3t}{t} \\ &= \lim_{t \rightarrow 0} \frac{\sin 3t}{t} \end{aligned}$$

SOLUTION:

- Because, $\lim_{t \rightarrow 0} \frac{\sin 3t}{t} = \frac{0}{0}$, let $h = 3t$ so that $t = \frac{h}{3}$. As t approaches 0, h also approaches 0. Thus,

SOLUTION: 

- $$\begin{aligned}\lim_{t \rightarrow 0} \frac{\sin 3t}{t} &= \lim_{h \rightarrow 0} \frac{\sin h}{\frac{h}{3}} \\ &= 3 \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= 3(1) \\ &= 3\end{aligned}$$